

Baire Space

Definition 1.0.1

A space X is said to be a **Baire space** if the following condition holds: Given any countable collection $\{A_n\}$ of closed sets of X each of which has empty interior in X , their union $\bigcup A_n$ also has empty interior in X .

Definition 1.0.2

A subset A of a space X was said to be of the **first category** in X if it was contained in the union of a countable collection of closed sets of X having empty interiors in X ; otherwise, it was said to be of the **second category** in X .

Theorem 1.0.3

A space X is a **Baire space** if and only if every nonempty open set in X is of the **second category**.